

A. 架設 DNS so easy !

1. 請解釋這四行的意義。

A: “.:1053” 定義一個 Server，在 port 1053 上接收所有網域的 DNS 查詢。

“forward . 8.8.8.8:53” 是一個轉發規則，將所有無法在本地解析的 DNS 查詢，轉發給 Google 的公共 DNS 伺服器。

“errors” 印出 error msg

“log” 印出收到的每一筆 DNS 查詢紀錄

2. 結果:

```
$ dig @127.0.0.1 -p 1053 dns.google.com

; <<>> DiG 9.18.39-0ubuntu0.22.04.3-Ubuntu <<>> @127.0.0.1 -p 1053 dns.google.com
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 47848
;; flags: qr rd ra; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
;; QUESTION SECTION:
;dns.google.com.          IN A

;; ANSWER SECTION:
dns.google.com.          900 IN A   8.8.8.8
dns.google.com.          900 IN A   8.8.4.4

;; Query time: 13 msec
;; SERVER: 127.0.0.1#1053(127.0.0.1) (UDP)
;; WHEN: Fri May 15 21:31:09 CST 2026
;; MSG SIZE rcvd: 103
```

3. 結果:

```
$ dig @127.0.0.1 -p 1053 dns.hinet.net

; <<>> DiG 9.18.39-0ubuntu0.22.04.3-Ubuntu <<>> @127.0.0.1 -p 1053 dns.hinet.net
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 32227
;; flags: qr rd ra; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
;; QUESTION SECTION:
;dns.hinet.net.          IN A

;; ANSWER SECTION:
dns.hinet.net.          21143 IN A   168.95.192.1
dns.hinet.net.          21143 IN A   168.95.1.1

;; Query time: 3 msec
;; SERVER: 127.0.0.1#1053(127.0.0.1) (UDP)
;; WHEN: Fri May 15 21:35:45 CST 2026
;; MSG SIZE rcvd: 100
```

B. 向全世界廣告你自己！

1. 如果我想要看到「hello www.nccu.lab」的話，我們應該下什麼 dig 指令呢？

A: dig @127.0.0.1 -p 1053 TXT www.cs.nccu.lab

```
$ dig @127.0.0.1 -p 1053 TXT www.cs.nccu.lab
; <<>> DiG 9.18.39-0ubuntu0.22.04.3-Ubuntu <<>> @127.0.0.1 -p 1053 TXT www.cs.nccu.lab
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 29165
;; flags: qr aa rd; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1
;; WARNING: recursion requested but not available

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: 40ecd865556ee020 (echoed)
;; QUESTION SECTION:
;www.cs.nccu.lab.      IN  TXT

;; ANSWER SECTION:
www.cs.nccu.lab.      3600  IN  TXT "hello www.nccu.lab"

;; Query time: 0 msec
;; SERVER: 127.0.0.1#1053(127.0.0.1) (UDP)
;; WHEN: Fri May 15 21:45:19 CST 2026
;; MSG SIZE rcvd: 102
```

2. 請試著理解「@」以及「www」的意義，然後更改 db.cs.nccu.lab，讓查詢 ds.cs.nccu.lab 的 TXT 時，會回傳「cf.liao nccu」，並將 db.cs.nccu.lab 的內容貼上

A: 「@」的意義：根網域。「www」的意義：子網域。

```
@ 3600 IN SOA sns.dns.icann.org. noc.dns.icann.org. (
    2017042745 ; serial
    7200       ; refresh (2 hours)
    3600       ; retry (1 hour)
    1209600    ; expire (2 weeks)
    3600       ; minimum (1 hour)
)
    3600 IN A 127.0.0.1
    3600 IN TXT "hello nccu.lab"
www 3600 IN A 127.0.0.1
    3600 IN TXT "hello www.nccu.lab"
ds 3600 IN TXT "cf.liao nccu" ←
```

```
$ dig @127.0.0.1 -p 1053 TXT ds.cs.nccu.lab
; <<>> DiG 9.18.39-0ubuntu0.22.04.3-Ubuntu <<>> @127.0.0.1 -p 1053 TXT ds.cs.nccu.lab
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 26510
;; flags: qr aa rd; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1
;; WARNING: recursion requested but not available

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: ea3aa860997c7b0b (echoed)
;; QUESTION SECTION:
;ds.cs.nccu.lab.      IN  TXT

;; ANSWER SECTION:
ds.cs.nccu.lab.      3600  IN  TXT "cf.liao nccu"

;; Query time: 0 msec
;; SERVER: 127.0.0.1#1053(127.0.0.1) (UDP)
;; WHEN: Fri May 15 21:56:19 CST 2026
;; MSG SIZE rcvd: 94
```

C. 逆向 DNS 查詢

A:

```
$ dig -x 140.124.183.43

; <<>> DiG 9.18.39-0ubuntu0.22.04.3-Ubuntu <<>> -x 140.124.183.43
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NXDOMAIN, id: 30044
;; flags: qr rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 1, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
;; QUESTION SECTION:
;43.183.124.140.in-addr.arpa.    IN PTR

;; AUTHORITY SECTION:
124.140.in-addr.arpa.    600 IN SOA galaxy.cc.ntut.edu.tw. network.ntut.edu.tw. 2018032356 10800 3600 2419200 900

;; Query time: 51 msec
;; SERVER: 10.255.255.254#53(10.255.255.254) (UDP)
;; WHEN: Fri May 15 22:00:55 CST 2026
;; MSG SIZE rcvd: 152
```

可以看到 ntut（國立臺北科技大學）是此 IP 的 Authoritative Name Server，至於這個 Authoritative Name Server 是誰提供的？流程是：Local -> Root -> TLD -> Authoritative，所以告訴我 AUTHORITY SECTION 的人是 TLD DNS。